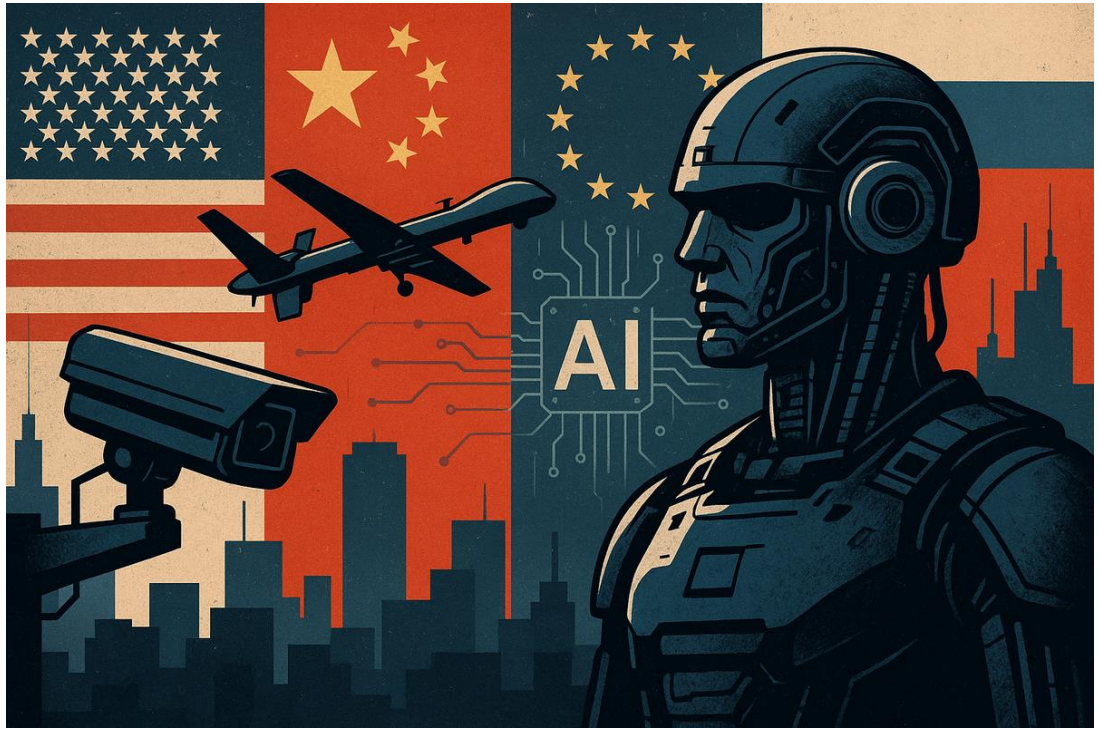
9 AI and Intellectual Property

GEB2508 AI and Society

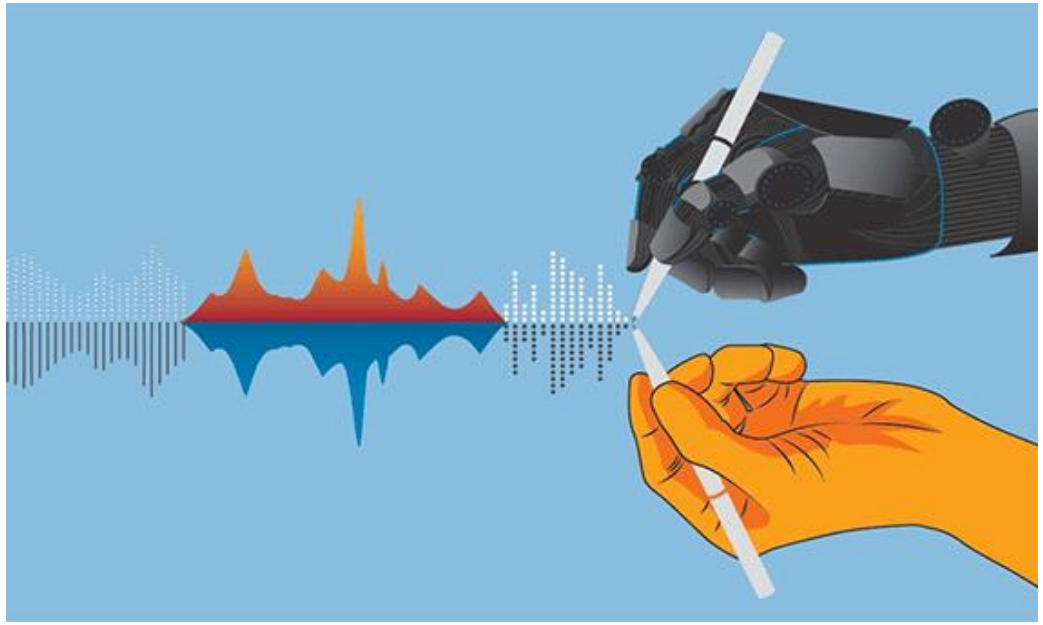
Review of Last Class

- Warfare and Technology
  - Cold War: Arms Race and Space Race
  - Military-Industrial Complex
  - Autonomous Weapons and Cyberwarfare
- Nation-State and Geopolitics
  - State Theory
  - Industrial Policy and Techno-nationalism
  - AI Arms Races and Geopolitics



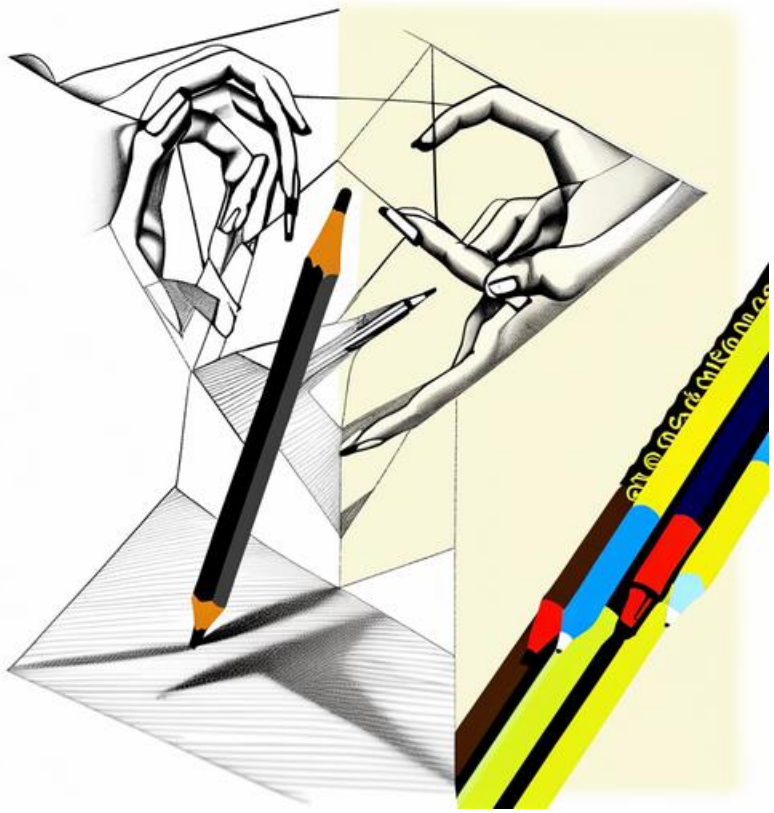
Content

- Authorship and Copyright
  - Authorship in the Age of AI
  - Copyright and Training Data
- Political Economy of Knowledge
  - Knowledge as A Public Good
  - Enclosure of The Commons
  - Open-Source vs. Proprietary AI
- Knowledge Governance Regimes
  - Technocracy and AI



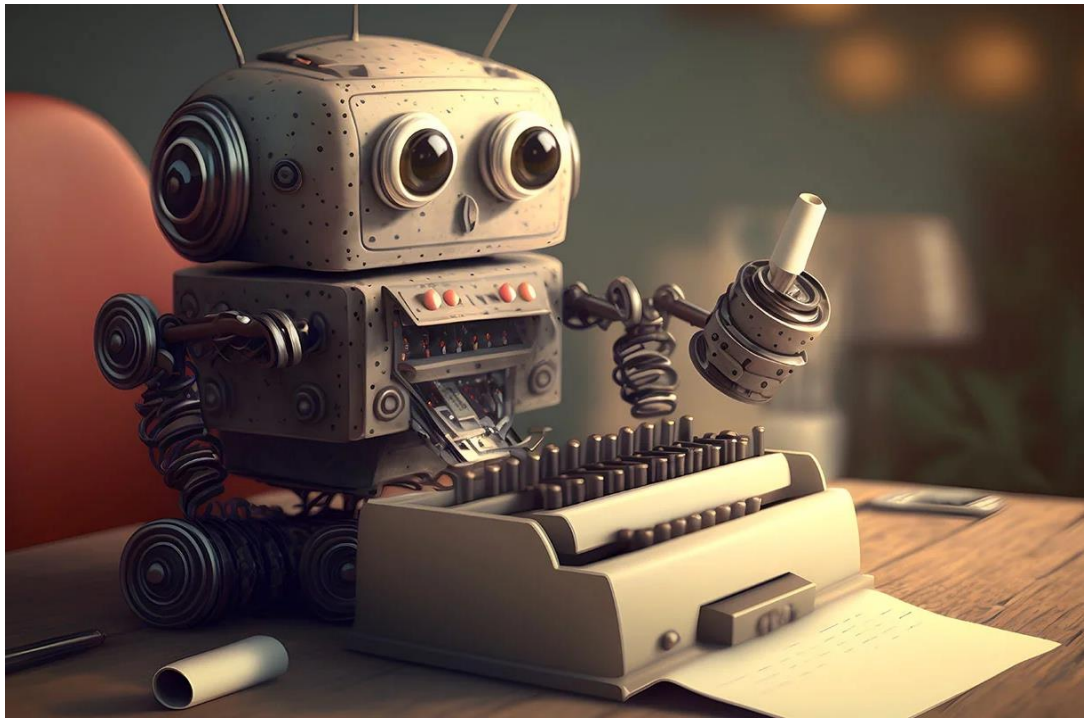
Philosophical Theories of Authorship

- Lockean **labor**
  - If a model generates a work autonomously, **whose** labor is rewarded?
- Hegelian **personality**
  - Strongest argument **against** AI authorship
- Utilitarianism
  - Would recognizing AI-generated works as ownable increase or decrease overall innovation?



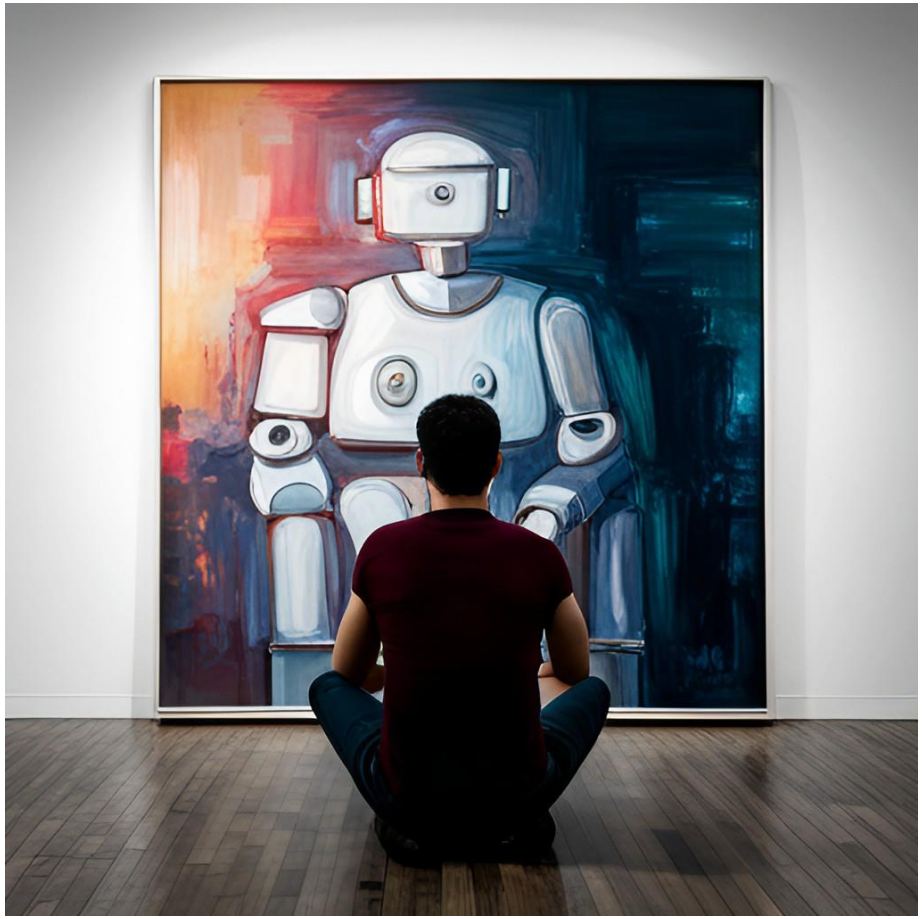
AI Authorship

- Courts rejected AI inventorship
  - Thaler v. Vidal; Thaler v. Comptroller-General of Patents
- **Anthropocentric** IP foundations
  - Hold property, transfer rights, and bear responsibility
- Human-AI collaboration
  - Standard of **meaningful** human control



Originality in the Age of AI

- Originality as copyright threshold
  - What counts as “original” in AI-generated works?
- Historical precedents: camera & Photoshop
  - Creativity located in **human choices**
- Originality as a **spectrum** rather than a binary
  - AI as a tool vs. AI creativity



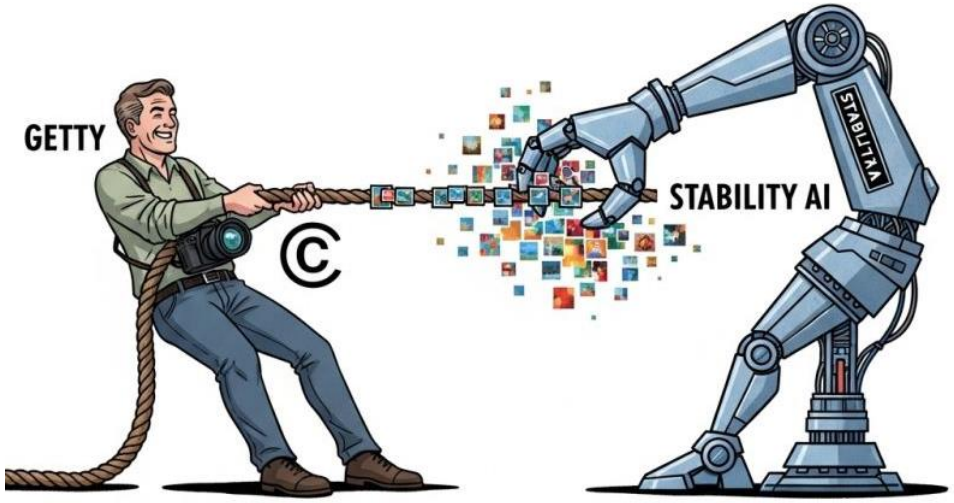
Copyright and AI Training

- Training on copyrighted material: **infringement** or **fair use**?
  - Derivative vs. transformative use
- Policy choice: **opt-in** (licensing before training) vs. **opt-out** (training by default)
  - Centralized vs. open
- Key lawsuits
  - The New York Times v. OpenAI
  - Getty Images v. Stability AI

The New York Times

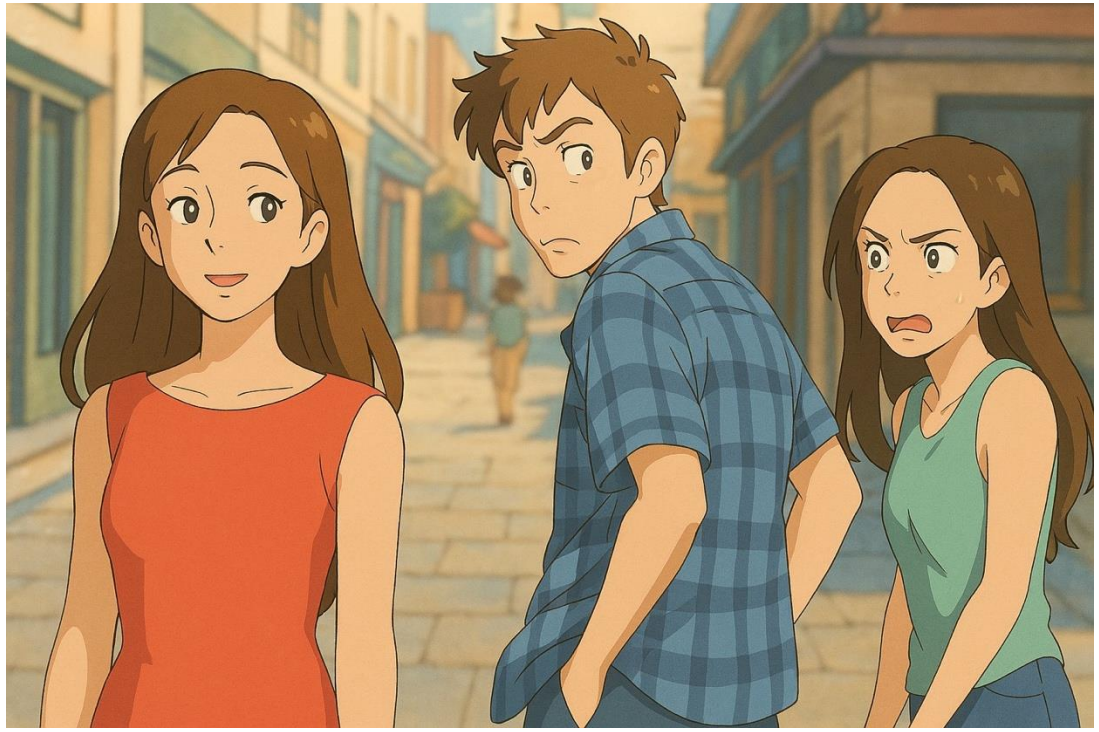
vs.

OpenAI



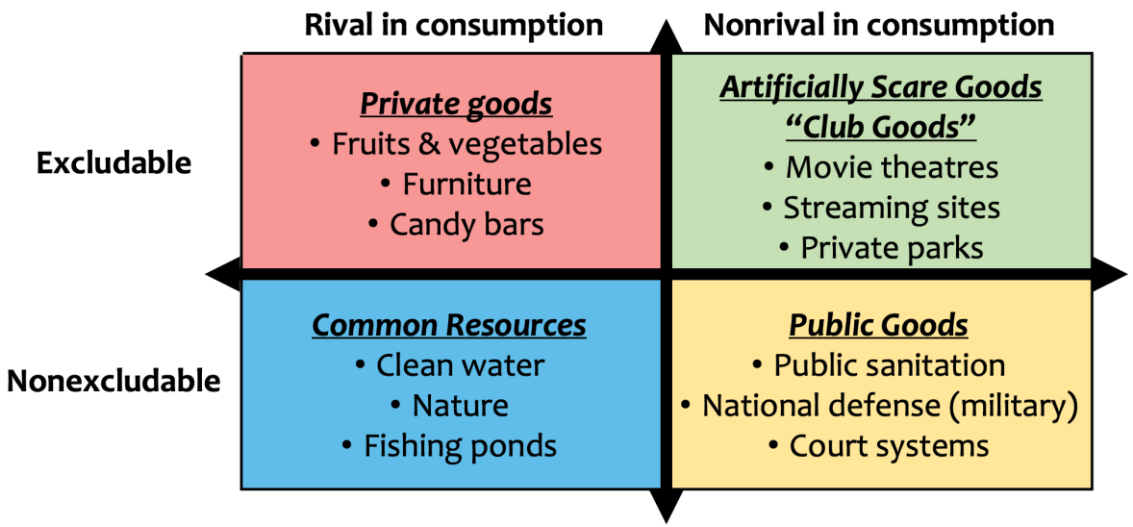
Style and Imitation

- Style vs. copyright: traditionally, style is not protected; AI makes styles reproducible **at scale**
- “In the style of …” prompts
  - Visual art
  - Writing and journalism
  - Music and voice cloning
- Style as **identity** marker
  - Personality rights



Private, Common, and Public Goods

- Private goods: rival and excludable
  - Ownership and **market** mechanisms
- Common goods: rival but non-excludable
  - Tragedy of the commons
- Public goods: non-rival and non-excludable



Knowledge as A Public Good

- Knowledge is a public, non-rival, and non-excludable good, so **market** mechanism alone cannot generate optimal outcomes.
- Benefits: widespread access and **positive externalities**
- Challenges
  - Free-rider problem
  - **Underinvestment**

Real Property vs. Intellectual Property

Real Property

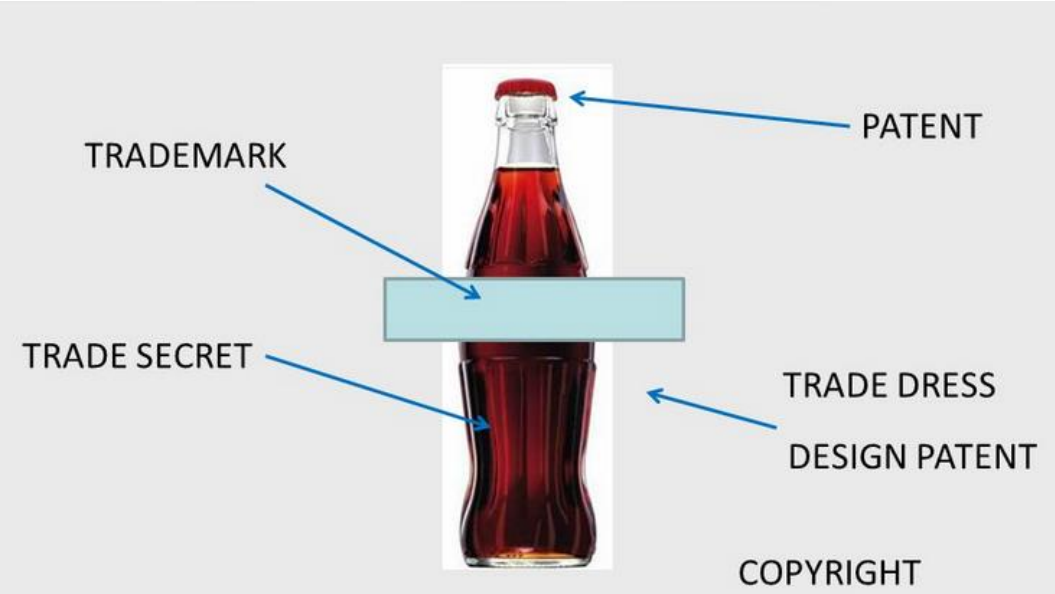
- **Tangible** assets that have a physical existence (e.g., land, food, vehicles)
- Rivalrous, scarce, and excludable
- Ownership limits: country-specific and no time limit

Intellectual Property

- **Intangible** assets that result from creative or intellectual efforts (e.g., inventions, artistic works)
- Non-rivalrous, non-scarce, partially excludable
- Ownership limits: country-specific and **time-limited**

Intellectual Property Rights (IPR)

- Schumpeterian innovation: economic progress is driven by entrepreneurial innovation.
- **Temporary monopolistic** profits incentivize firms to invest in research and development.
  - The privilege is a means to an end, not an end in itself.
- Patents, trademarks, copyrights

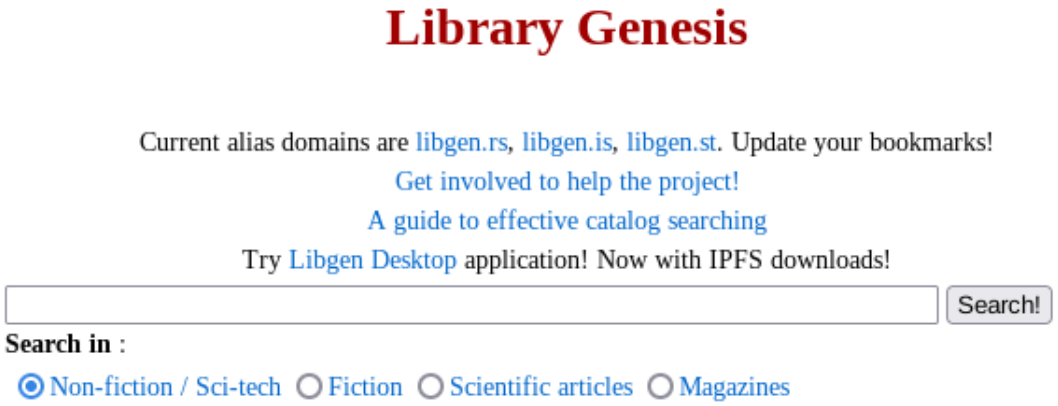


Enclosure of The Commons

- Enclosure: **privatize publicly-owned** commons by assigning property rights to exclusive use
  - “A revolution of the rich against the poor” (Karl Polanyi)
- Tragedy of the commons & tragedy of the anticommons
  - **Overuse** of natural resource commons & **underuse** of knowledge commons

The Case of SciHub and LibGen

- Universities in developing countries excluded from the global scientific community
- **Taxpayer-funded** research locked behind paywalls
- Publishers profit **disproportionately** while contributing little to the actual research process.
- Knowledge equity



The Case of Fandom Creativity

- Creativity as transformation, not originality: blurring producer vs. consumer
- **Non-market** value system: recognition, community, and participation
- **Knowledge commons** in practice: shared tropes, tags, and collaborative world-building

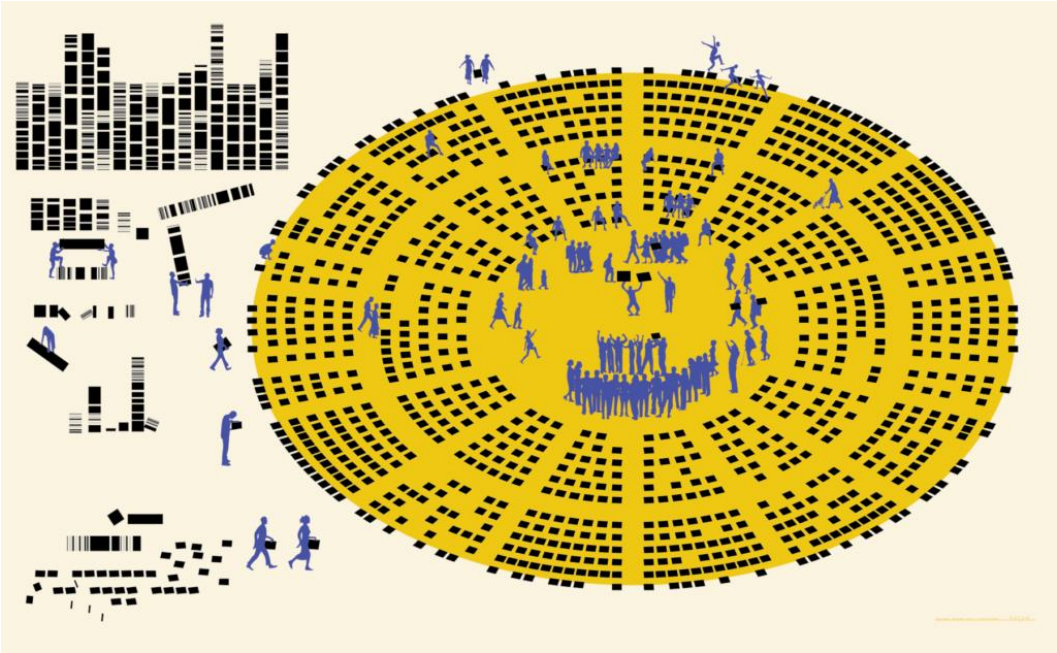
Archive of Our Own

Public Domain and Knowledge Commons

- Public domain
  - Expired copyright (author life + 70 years)
  - Works never eligible for protection (facts, ideas, style)
- Functions: access to knowledge; raw material for new creativity
- Public domain as AI training **infrastructure**
- Reliance on public domain vs. enclosure of proprietary datasets

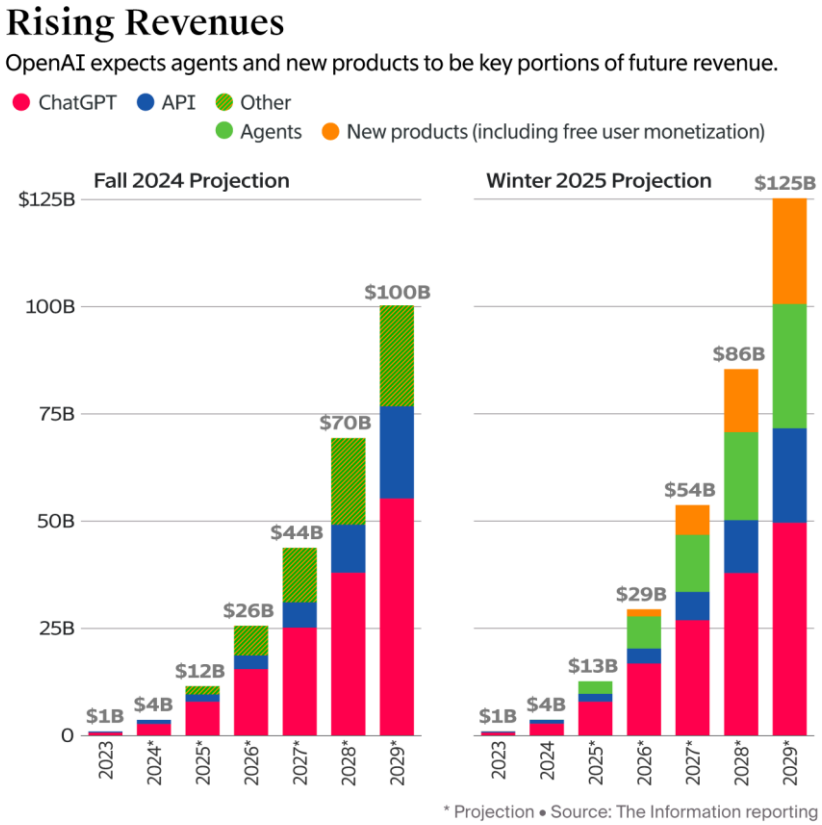
Digital Commons

- Digital commons (ideal): decentralization, peer production, open-source software
- Enclosure: from open knowledge to **proprietary** platforms & datasets
  - User-generated content as free raw material
  - Asymmetry between **collective** input and **private** profit
- Transformation in the **governance** of knowledge



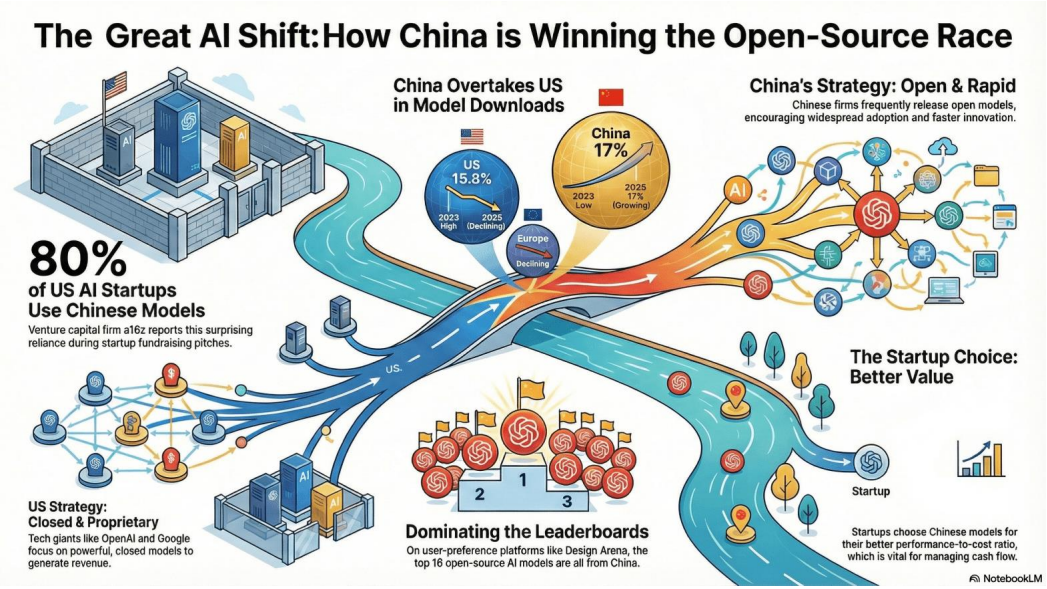
US Proprietary AI Model

- Closed and tightly controlled ecosystems
- API pricing → AI as a **metered utility**
- IP as barrier to entry → **self-reinforcing scale**
- High switching costs → platform **dependency**
- US venture-capital-driven tech development: seeking **monopoly** rents



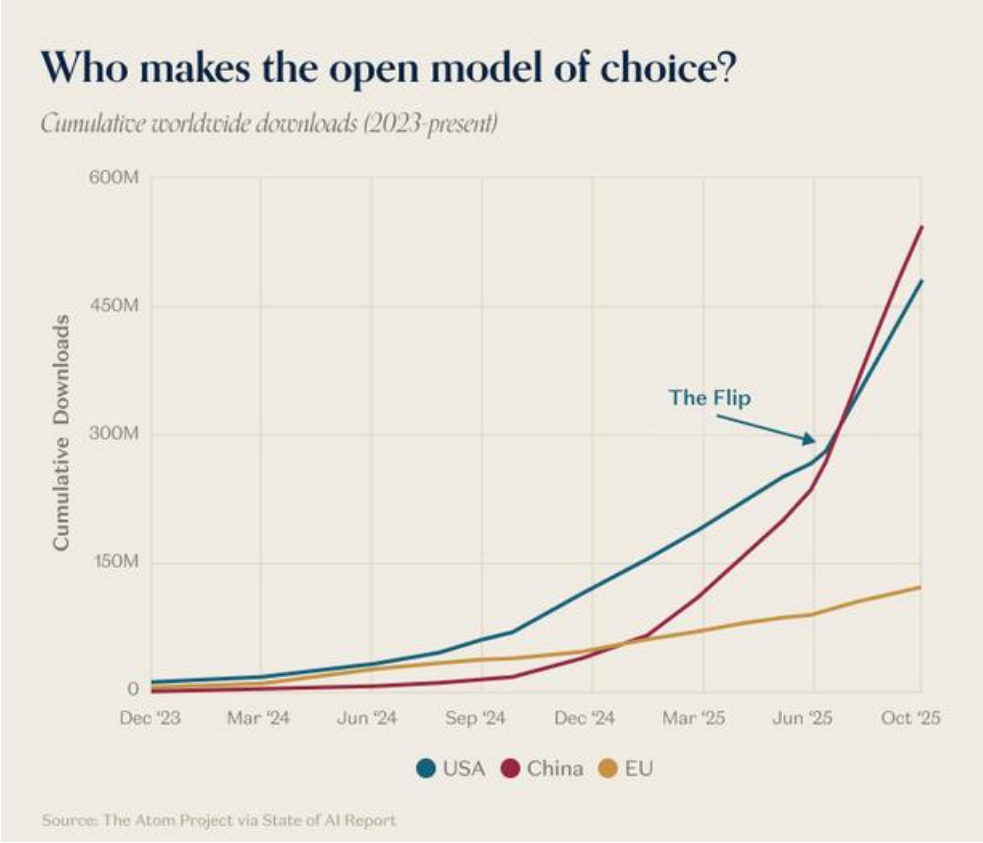
China’s Strategic Pivot to Open Models

- Shaped by state coordination, industrial policy, and goals of technological self-sufficiency
- Lower-cost deployment, rapid diffusion and sectoral adaptation
- Geopolitical dynamics: U.S. semiconductor restrictions; **data-sovereignty** concerns
- Openness as **infrastructural public good** (nationally bounded)



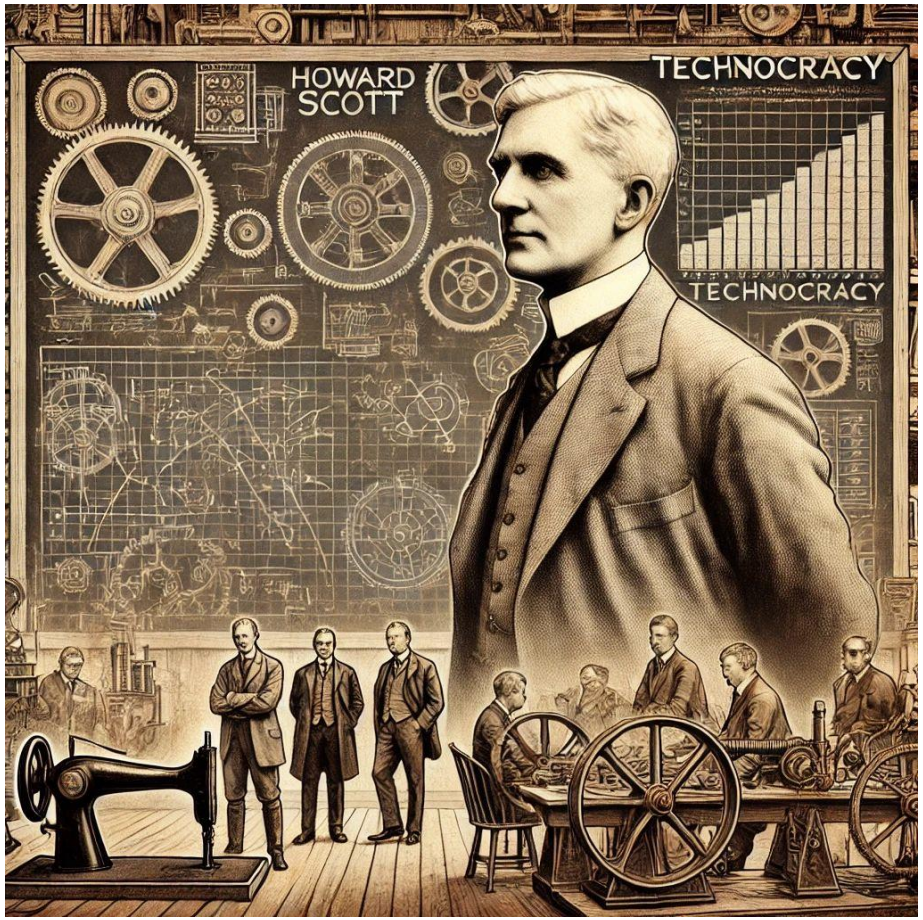
Open-Source vs. Proprietary AI

- Beyond technical or ideological: competing **innovation regimes** and geopolitical strategy
  - Venture-capital, platform-based system vs. AI as national infrastructure in a state-driven system
- Reframing global competition: which regime can **scale** and **diffuse** effectively across diverse contexts
- Control shifts from ownership of **models** to control of **ecosystems**.



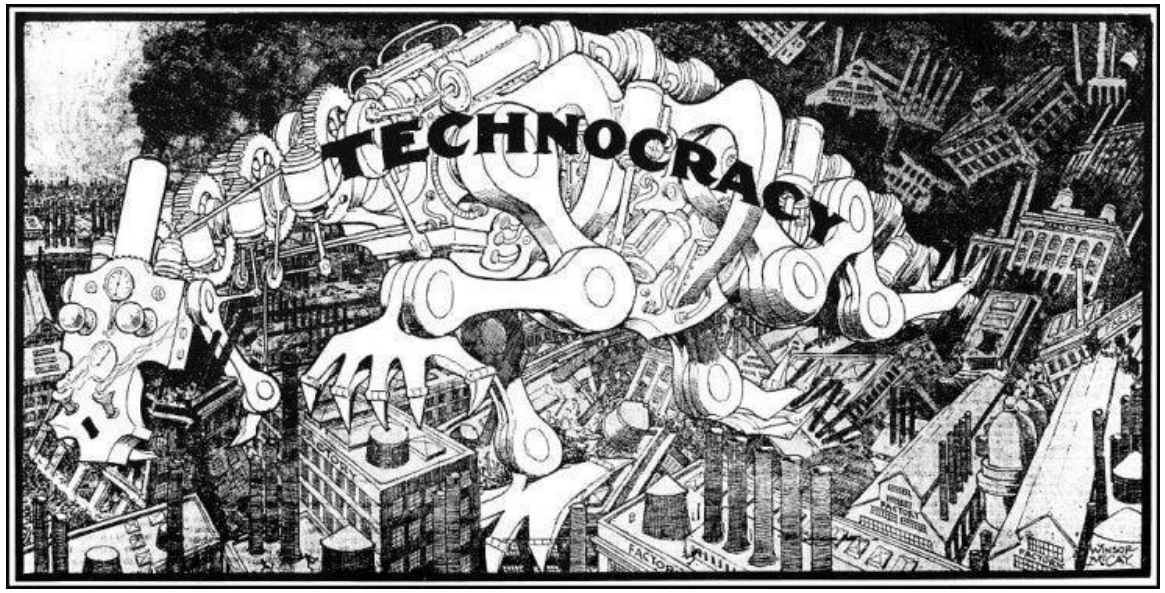
Technocracy: Historical Origins

- Utopian socialism (1820s-1840s)
  - Saint-Simon: governments replaced by industrial and scientific councils
  - Centralized planning
- Technocracy movement in the US (1920s–1930s)
  - Howard Scott and Thorstein Veblen
  - Emphasis on using science and data to manage economies and societies



Principles and Critiques

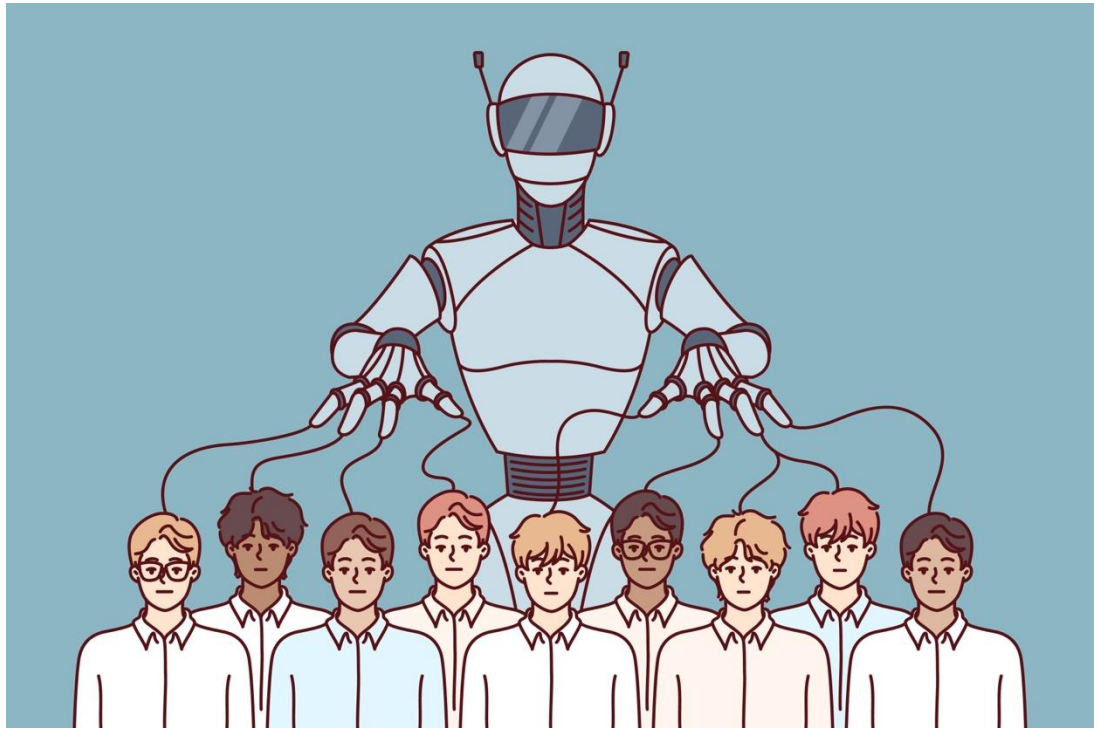
- Rule by experts
- Data-driven decision-making
- Rationality and efficiency
- Historical examples: Soviet Union, the West, China
- Critics: democratic principles, transparency and accountability, elitism, and exclusion



Technocracy in Dystopian Fiction

1984

Brave New World

Technocracy in the Age of AI

- From Cold War systems analysis & cybernetics to algorithmic governance
- Technocratic AI and **legitimacy** problems: black-box decision making; lack of contestability; procedural justice
- Is AI governance becoming a new technocracy?
- Technocratic efficiency vs. democratic legitimacy